Project Proposal

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2026-03-15

Which assumption—discrete regime shifts (HMM) or continuous volatility evolution (Stochastic Volatility)-better explains the market dynamics during the Dot-com bubble, the 2008 Financial Crisis and the COVID-19 crisis?

- ^GSPC (S&P 500 Index): Top 500 stocks in the U.S. (probably should use this).
- ^VIX (CBOE Volatility Index): Market volatility measure that we can use for model validation.
- XLK (Technology Select Sector SPDR Fund): An ETF containing all the tech stocks in the S&P 500. Maybe use for dot com crash?
- XLF (Financial Select Sector SPDR Fund): An ETF containing all the financial stocks in the S&P 500 (banks, insurance, etc.). Check for 2008 Financial Crisis?
- TLT (iShares 20+ Year Treasury Bond ETF): This tracks long-term US government bonds. "Safe haven" asset that normally behaves opposite to stocks. Does the model still work with this IDK?

We use log return $log(P_t/P_{t-1})$ because it makes it easier to work with financial data.

```
project_tickers <- c("^GSPC", "^VIX", "XLK", "XLF", "TLT")
market_data <- tq_get(project_tickers,
                      get  = "stock.prices",
                      from = "1998-01-01")
```

```
## Warning: There were 5 warnings in `dplyr::mutate()`.
## The first warning was:
## i In argument: `data.. = purrr::map(...)`.
## Caused by warning in `zoo()`:
## ! some methods for "zoo" objects do not work if the index entries in 'order.by' are not unique
## i Run `dplyr::last_dplyr_warnings()` to see the 4 remaining warnings.
```

```
weekly_returns <- market_data %>%
  mutate(week = floor_date(date, "week")) %>%
  group_by(symbol, week) %>%
  arrange(date) %>%
  slice_tail(n = 1) %>%
  ungroup()


weekyly_log_returns <- weekly_returns %>%
  group_by(symbol) %>%
  arrange(date) %>%
```

```r
  mutate(
    previous_price = lag(adjusted),
    log_return = log(adjusted / previous_price)
  ) %>%

  filter(!is.na(log_return)) %>%
  select(symbol, date, adjusted, log_return)

head(weekyly_log_returns)
```

```
## # A tibble: 6 x 4
## # Groups:   symbol [2]
##   symbol date       adjusted log_return
##   <chr>  <date>        <dbl>      <dbl>
## 1 ^GSPC  1998-01-09    928.    -0.0498
## 2 ^VIX   1998-01-09     28.7    0.203
## 3 ^GSPC  1998-01-16    962.     0.0358
## 4 ^VIX   1998-01-16     21.6   -0.282
## 5 ^GSPC  1998-01-23    958.    -0.00409
## 6 ^VIX   1998-01-23     23.3    0.0743
```

Hidden Markov Model

The Likelihood

Let Y_t represent the weekly log return of the index at time t, and let $X_t \in \{1, 2\}$. State 1 represents a "Calm/Normal" market regime, and State 2 represents a "Crisis" regime.

$$Y_t \mid X_t = i \sim \mathcal{N}(\mu_i, \sigma_i^2)$$

Where:

$$\mu_1, \mu_2 \sim \mathcal{N}(0, 1)$$
$$\sigma_1, \sigma_2 \sim \text{Exponential}(1)$$

(Note: Should we enforce $\sigma_1 < \sigma_2$, ensuring State 1 always represents the lower-volatility normal market, and State 2 always represents the crisis?). 2. The Latent State Dynamics (The Regime Switching)The sequence of unobserved market states follows a first-order Markov process. The probability distribution of tomorrow's state, X_t, is explicitly conditioned on today's state, X_{t-1}.

Latent States

$$X_t \mid X_{t-1} = 1 \sim \text{Categorical}(P_{11}, 1 - P_{11})$$
$$X_t \mid X_{t-1} = 2 \sim \text{Categorical}(1 - P_{22}, P_{22})$$

Transition Probabilities

$$P = \begin{pmatrix} P_{11} & 1 - P_{11} \\ 1 - P_{22} & P_{22} \end{pmatrix}$$

where,

$$P_{11} \sim \text{Beta}(1, 1)$$
$$P_{22} \sim \text{Beta}(1, 1)$$